

Higher Order DE

$$a_0 \frac{d^ny}{dx^n} + a_1 \frac{d^{n-1}y}{dx^{n-1}} + \dots + a_n y = X \quad \text{--- (1)}$$

$a_0, a_1, \dots, a_n$ are const. & X is funⁿ of x .

this form is called as L.D.Eqⁿ with const. coeff.

Consider $\frac{d}{dx} = D$ $\frac{d}{dx} y \neq y \frac{d}{dx}$

from (1)
 $a_0 D^n y + a_1 D^{n-1} y + \dots + a_n y = X$

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_n) y = X$$

$$f(D) y = X \quad \text{--- (2)}$$

solⁿ of (2) is called as General solⁿ

General solⁿ = Complementary Function + Particular Integral.

$$y = C.F + P.I \quad / \quad y = y_c + y_p$$

(1) C.F. from (2) $f(D)y = X$

consider $f(D)y = 0$

Here, $f(D)=0$ is called Auxiliary Eqⁿ.

We get Complementary funⁿ by finding roots of Auxiliary Eqⁿ. using following methods

Case (1) If roots are real & distinct.
 say m_1, m_2, m_3

$$y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

(2) If roots are real & repeated.

Suppose root a is repeated thrice

then
 $y_c = C_1 e^{ax} + C_2 x e^{ax} + C_3 x^2 e^{ax}$

eg. suppose roots are 3, 3, -2

$$y_c = C_1 e^{3x} + C_2 x e^{3x} + C_3 e^{-2x}$$

$$e^{i\theta} = \cos\theta + i \sin\theta$$

(3) If roots are complex say $a \pm ib$

$$y_c = e^{ax} (C_1 \cos bx + C_2 \sin bx)$$

$$\begin{aligned} & e^{(a+ib)x} \\ & e^{ax} e^{ibx} \\ & e^{ax} (\cos bx + i \sin bx) \end{aligned}$$

Suppose roots are $a \pm ib, a \pm ib$ (repeated complex root)

$$y_c = e^{ax} (C_1 \cos bx + C_2 \sin bx) + x e^{ax} (C_3 \cos bx + C_4 \sin bx)$$

$$= e^{ax} [(C_1 + x C_3) \cos bx + (C_2 + x C_4) \sin bx]$$

$$f(D)y = x \quad (\quad)$$

$$G \cdot S = C \cdot f + P \cdot Z$$

$$y = y_c + y_p$$

$$f(D)y = 0$$

$$f(D) = 0$$

real & distinct

m_1, m_2

$$y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

real same

m, m

$$y_c = C_1 e^{mx} + C_2 x e^{mx}$$

Complex

$a \pm i b$

$$y_c = e^{ax} \left(C_1 \cos bx + C_2 \sin bx \right)$$

$$\diamond 9 \frac{d^2 y}{dx^2} + 18 \frac{dy}{dx} - 16y = 0$$

$$D = \frac{d}{dx}$$

$$9D^2 y + 18Dy - 16y = 0$$

$$(9D^2 + 18D - 16)y = 0$$

$$\text{Auxiliary eq}^n \quad 9D^2 + 18D - 16 = 0$$

$$D = \frac{2}{3}, -\frac{8}{3}$$

$$y = C_1 e^{\frac{2}{3}x} + C_2 e^{-\frac{8}{3}x}$$

$$\diamond \frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 6y = 0$$

$$D = \frac{d}{dx}$$

$$D^3y + 2D^2y - 5Dy - 6y = 0$$

$$(D^3 + 2D^2 - 5D - 6)y = 0$$

$$\text{A.E.V} \quad D^3 + 2D^2 - 5D - 6 = 0$$

$$D = 2, -1, -3$$

$$y = C_1 e^{2x} + C_2 e^{-x} + C_3 e^{-3x} //$$

$$\diamond (D^4 - 5D^3 + 5D^2 + 5D - 6)y = 0$$

A eqn $D^4 - 5D^3 + 5D^2 + 5D - 6 = 0$

$$D = 3, 2, 1, -1$$

$$y = C_1 e^{3x} + C_2 e^{2x} + C_3 e^x + C_4 e^{-x}.$$

$$\diamond \frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4y = 0$$

hw

$$(D-1)^4 y = 0$$

$$D^3 y - 3D^2 y + 4y = 0$$

$$(D^3 - 3D^2 + 4)y = 0$$

$$A - 2q^n \quad D^3 - 3D^2 + 4 = 0$$

$$D = -1, 2, 2$$

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 x e^{2x}$$

$$\diamond \frac{d^4 y}{dx^4} - 18 \frac{d^2 y}{dx^2} + 81y = 0$$

$$D = \frac{d}{dx}$$

$$D^4 y - 18D^2 y + 81y = 0$$

$$(D^4 - 18D^2 + 81)y = 0$$

$$A. \text{ Eqn } D^4 - 18D^2 + 81 = 0$$

$$D = 3, 3, -3, -3,$$

$$y = C_1 e^{3x} + C_2 x e^{3x} + C_3 e^{-3x} + C_4 x e^{-3x} //$$

$$D^2 = m$$

$$m^2 - 18m + 81 = 0$$

$$m = 9, 9$$

$$D^2 = 9, 9$$

$$D = \pm 3, \pm 3$$

❖ Solve $\frac{d^3 y}{dx^3} + 8y = 0$

$$D^3 y + 8y = 0$$

$$(D^3 + 8)y = 0$$

A e^{ax} , $D^3 + 8 = 0$

$$D = -2, 1 \pm i\sqrt{3}$$

$$a = 1, b = \sqrt{3}$$

$$y = C_1 e^{-2x} + e^x (C_2 \cos \sqrt{3}x + C_3 \sin \sqrt{3}x)$$

$$a \pm ib$$

$$e^{ax} (C_1 \cos bx + C_2 \sin bx)$$

$$\diamond (D^2 - 2D + 2)^2 y = 0$$

$$a \pm ib$$

$$A e^{ax} \quad (\underline{D^2 - 2D + 2})^2 = 0$$

$$D = 1 \pm i, 1 \pm i$$

$$a=1, b=1$$

$$y = e^x (c_1 \cos x + c_2 \sin x) + x e^x (c_3 \cos x + c_4 \sin x)$$

$$\diamond (D^4 + 4)y = 0$$

$$D^4 + 4 = 0$$

$$D = \underline{\underline{1 \pm i}}, \quad \underline{-1 \pm i}$$

$$y = e^x (c_1 \cos x + c_2 \sin x) + e^{-x} (c_3 \cos x + c_4 \sin x)$$

$$\diamond (D^4 + 8D^2 + 16)y = 0$$

$$D^4 + 8D^2 + 16 = 0 \quad \checkmark$$

$$D^2 = m$$

$$m^2 + 8m + 16 = 0$$

$$m = -4, -4$$

$$D^2 = -4, -4$$

$$D = \pm \sqrt{-4}, \pm \sqrt{-4}$$

$$= \pm 2i, \pm 2i$$

$$a = 0, b = 2$$

$$y = C_1 \cos 2x + C_2 \sin 2x + x(C_3 \cos 2x + C_4 \sin 2x)$$

$$\diamond \{(D^2 + 1)^3 (D^2 + D + 1)^2\} y = 0$$

A eqn $\underline{(D^2 + 1)^3} (D^2 + D + 1)^2 = 0$

$$D = \underbrace{\pm i, \pm i, \pm i}_{a=0, b=1}, \underbrace{-\frac{1}{2} \pm i \frac{\sqrt{3}}{2}, -\frac{1}{2} \pm i \frac{\sqrt{3}}{2}}_{a=-\frac{1}{2}, b=\frac{\sqrt{3}}{2}}$$

$$y = (C_1 \cos x + C_2 \sin x) + x (C_3 \cos x + C_4 \sin x) \\ + x^2 (C_5 \cos x + C_6 \sin x) + e^{-\frac{1}{2}x} (C_7 \cos \frac{\sqrt{3}}{2}x \\ + C_8 \sin \frac{\sqrt{3}}{2}x) + x e^{-\frac{1}{2}x} (C_9 \cos \frac{\sqrt{3}}{2}x + C_{10} \sin \frac{\sqrt{3}}{2}x)$$